TAPH ESP32 CARRIER — REV 0 PRELIMINARY DESIGN-REVIEW PACKAGE

STATUS: NOT FOR FABRICATION, ASSEMBLY OR ENERGIZATION. Prepared for adviser/consultant review (2026-11-08).
Passing ERC/DRC is a rule-consistency check only; it is not evidence of electrical or operational safety.

ARCHITECTURE (rev 0)
- Carrier PCB for a SOCKETED ESP32 development board (provisional: Espressif ESP32-DevKitC V4, classic ESP32, 38-pin).
- Three externally supplied low-voltage DC inputs, each a separate removable isolation point:
  VIN_LOGIC (J1, 12 V nominal) -> protection -> 5 V module -> +5V / DevKitC 5V
  PUMP_VIN (J2, 12 V nominal) -> fuse -> pump connector, low-side MOSFET switch with hardware default-OFF
  SERVO_VIN (J4, per servo) -> fuse -> servo connector; PWM buffered 3.3 V -> 5 V
- No mains voltage on the PCB. Generator power does not pass through the PCB (sense connections only).
- Measurement: generator voltage sense + external generator current sensor interface (analog or I2C),
  pump current (on-board shunt + INA180) and pump supply voltage. Flow pulse input; optional RPM and pitch feedback.
- OLED on I2C (IO21/IO22), I2C expansion, spare GPIO, test points on every rail and key signal.

FIELD "Status" ON EVERY PART: IMPLEMENTED / PROVISIONAL / TBD / OPTIONAL / DNP / CONFIRM (see pcb/rev0/README.md).
Values shown as "TBD" are intentionally unresolved: they depend on unverified external component ratings.

Logic power

Logic power input, protection and +5 V regulation

File: power.kicad_sch

ESP32 DevKitC socket

ESP32-DevKitC V4 socket carrier and GPIO allocation

File: mcu.kicad_sch

Pump switch

12 V pump supply, default-off low-side switch, current & voltage sensing

File: pump.kicad_sch

Servo interface

Blade-pitch servo: separate supply input and buffered PWM

File: servo.kicad_sch

Sensor inputs

Flow pulse, optional RPM pulse and optional pitch feedback

File: sensors.kicad_sch

Generator measurement

Generator voltage sense and current-sensor interface

File: genmeas.kicad_sch

HMI, expansion, test

OLED, I2C expansion, spare GPIO, test points, mounting

File: hmi.kicad_sch

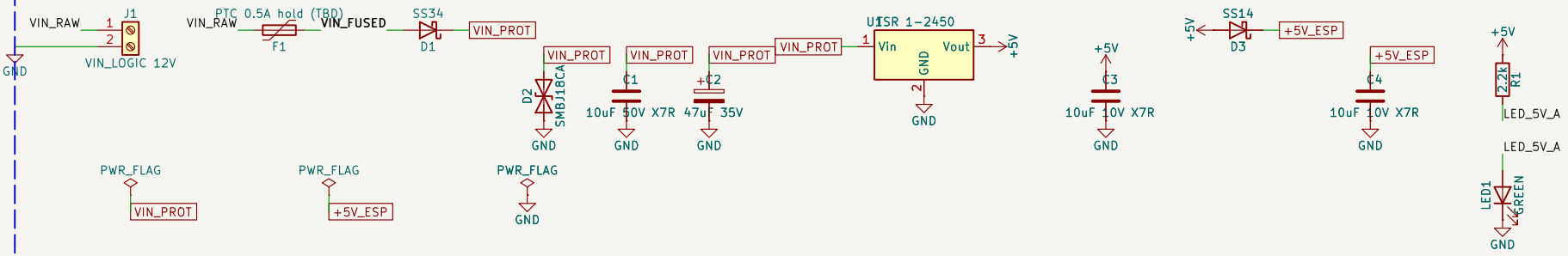
Generated by pcb/rev0/tools/gen_sch.py from design.py
Unverified external components: sensor, generator, pump, servo, supply. See pcb/rev0/docs.
Design-review prototype. Not reviewed for manufacture, assembly or energization.
TAPH research — PRELIMINARY — NOT FOR FABRICATION

Sheet: /
File: taph_carrier_rev0.kicad_sch

Title: TAPH carrier rev 0 — Architecture

Size: A3	Date: 2026-11-07	**Rev: 0-prelim**
KiCad E.D.A. 10.0.7		Id: 1/8

LOGIC SUPPLY (VIN_LOGIC 12 V nominal, external SELV supply — rating unverified)



VIN_RAW -> F1 -> D1 (reverse polarity) -> D2 TVS + C1/C2 -> U1 (5 V module) -> +5V rail.
+5V feeds: servo buffer U4 VCC, optional 5 V sensor supplies (via solder jumpers), D3 -> DevKitC 5V pin (+5V_ESP).
+3V3 is NOT generated on the carrier: it is the DevKitC on-board regulator output (capacity unverified; keep peripheral load small).
PWR_FLAGs mark nets that are genuinely powered from connectors/diodes (ERC driver annotation only).

UNRESOLVED: supply source, max/min voltage, battery/charger presence, logic load current, PTC rating.

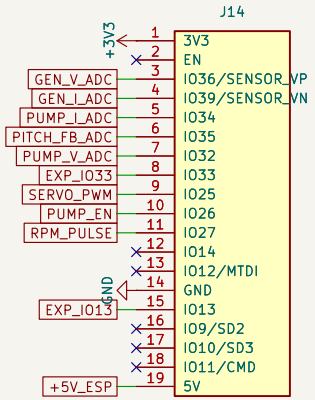
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Sheet: /Logic power/
File: power.kicad_sch

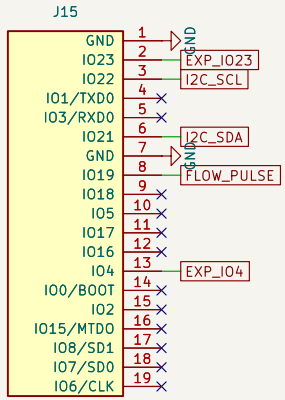
Title: TAPH carrier rev 0 — Logic power

Size: A4	Date: 2026-11-07	**Rev: 0-prelim**
KiCad E.D.A. 10.0.7		Id: 2/8

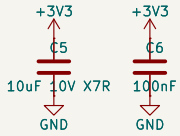
SOCKETED ESP32 DEV BOARD (provisional target: Espressif ESP32-DevKitC V4, 38-pin, classic ESP32) — CONFIRM BOARD, PINOUT AND ROW SPACING



DevKitC J2 socket



DevKitC J3 socket



PROPOSED GPIO MAP (rev 0 proposal — NOT evidence of historical wiring)
IO36 GEN_V_ADC   ADC1_CH0 input-only
IO39 GEN_I_ADC   ADC1_CH3 input-only
IO34 PUMP_I_ADC   ADC1_CH6 input-only
IO35 PITCH_FB_ADC   ADC1_CH7 input-only (optional)
IO32 PUMP_V_ADC   ADC1_CH4
IO19 FLOW_PULSE   interrupt input
IO27 RPM_PULSE   interrupt input (optional)
IO25 SERVO_PWM   LEDC PWM -> U4 buffer
IO26 PUMP_EN   Q1 gate drive (HW pull-down)
IO21 I2C_SDA   OLED + I2C expansion
IO22 I2C_SCL
IO4/13/23/33 EXP_* uncommitted expansion
Not used: IO0/2/5/12/15 (strapping), IO6-11 (flash),
IO1/3 (USB-UART), IO14 (boot-time PWM), IO16/17 (PSRAM
on WROVER variants), IO18, EN.
All analog inputs on ADC1 (ADC2 unusable with Wi-Fi).

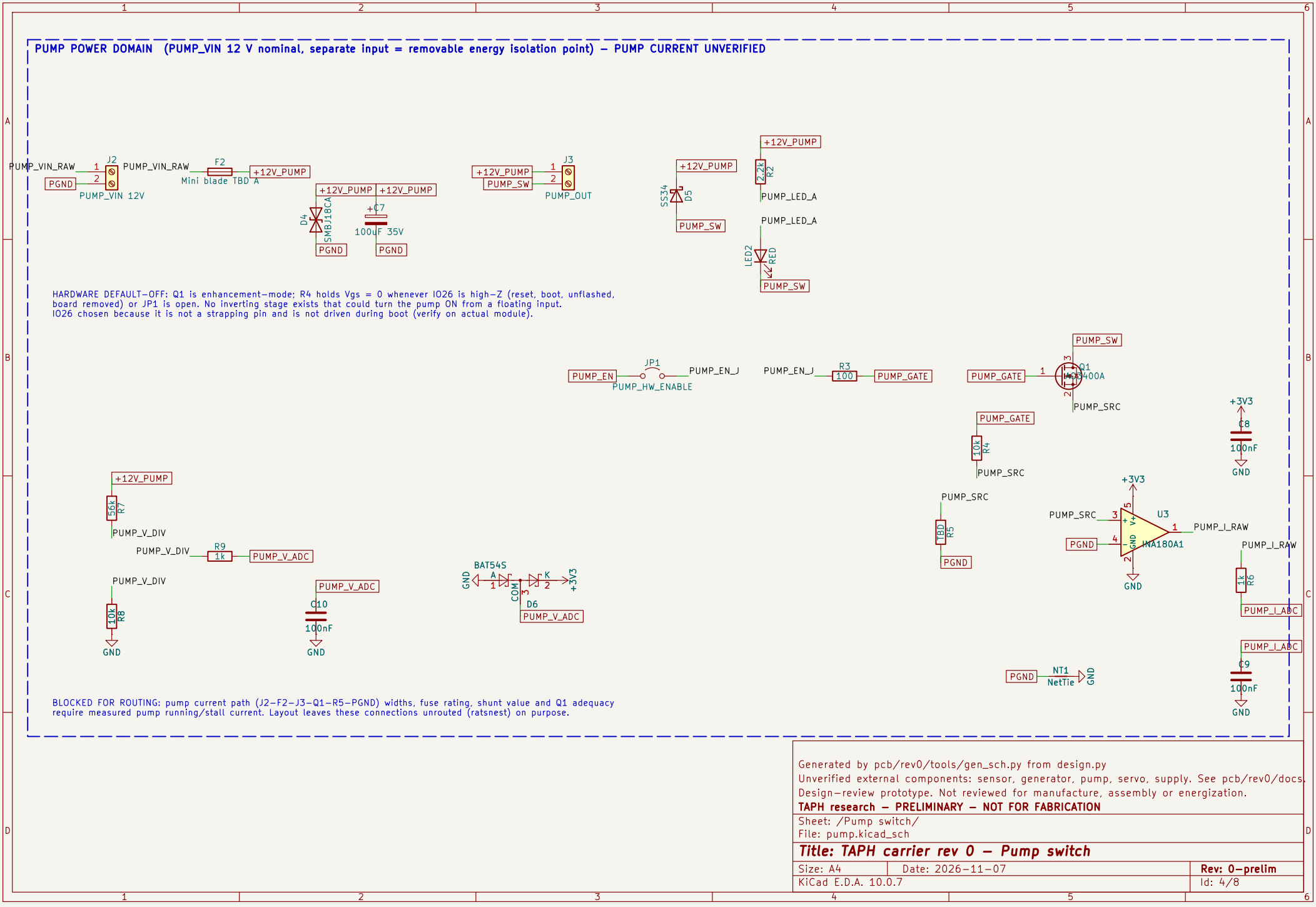
Historical sketch used OLED on IO2/IO0 (strapping pins), servo on IO5 (strapping) and fed a 5 V ACS712 output directly to IO34. Rev 0 deliberately moves I2C to IO21/22, servo to IO25 and conditions every analog input. Firmware local_config pins must be updated to this map after board confirmation.

Generated by pcb/rev0/tools/gen_sch.py from design.py
Unverified external components: sensor, generator, pump, servo, supply. See pcb/rev0/docs.
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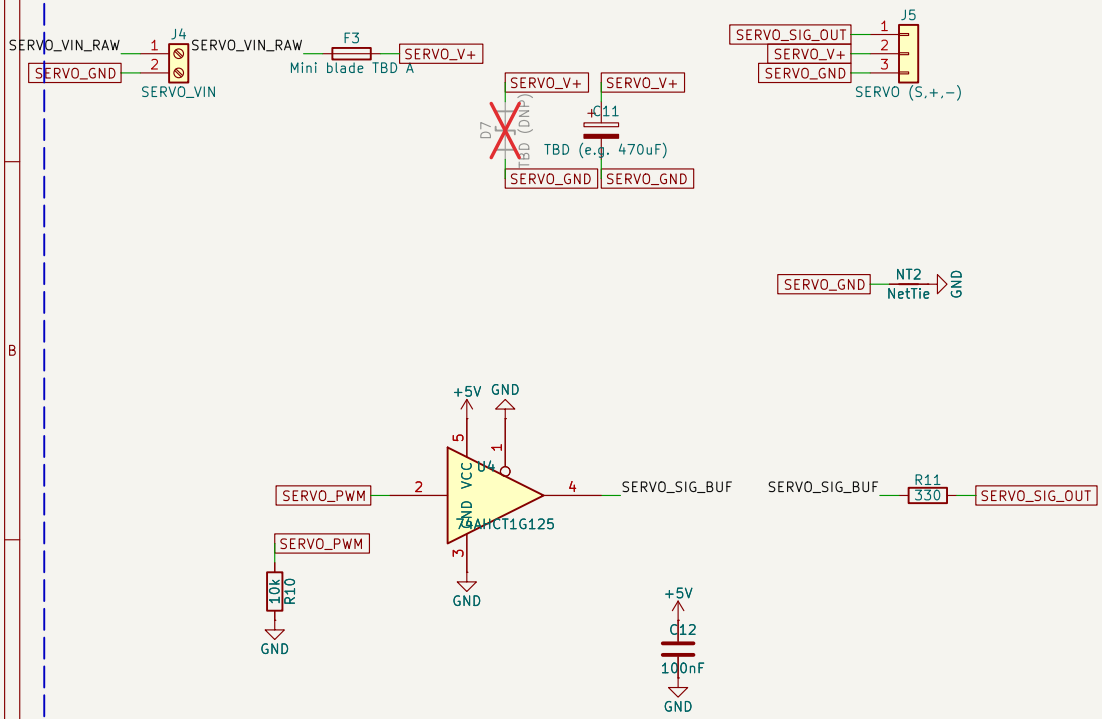
Sheet: /ESP32 DevKitC socket/
File: mcu.kicad_sch

Title: TAPH carrier rev 0 — ESP32 DevKitC socket

Size: A4	Date: 2026-11-07	**Rev: 0-prelim**
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PUMP POWER DOMAIN (PUMP_VIN 12 V nominal, separate input = removable energy isolation point) — PUMP CURRENT UNVERIFIED
PUMP_VIN_RAW
J2
PGND
PUMP_VIN 12V
F2
Mini blade TBD A
+12V_PUMP
D4
SMBJ18CA
+C7
100uF 35V
J3
PUMP_SW
PUMP_OUT
SS34
D5
2.2k
R2
PUMP_LED_A
LED2
RED
HARDWARE DEFAULT—OFF: Q1 is enhancement—mode; R4 holds Vgs = 0 whenever IO26 is high—Z (reset, boot, unflashed, board removed) or JP1 is open. No inverting stage exists that could turn the pump ON from a floating input.
IO26 chosen because it is not a strapping pin and is not driven during boot (verify on actual module).
PUMP_EN
JP1
PUMP_HW_ENABLE
PUMP_EN_J
R3
100
PUMP_GATE
Q1
PUMP_SRC
R4
10k
+3V3
C8
100nF
GND
56k
R7
PUMP_V_DIV
R9
1k
PUMP_V_ADC
R8
C10
BAT54S
D6
COM
R5
TBD
U3
INA180A1
PUMP_I_RAW
R6
PUMP_I_ADC
C9
NT1
NetTie
BLOCKED FOR ROUTING: pump current path (J2—F2—J3—Q1—R5—PGND) widths, fuse rating, shunt value and Q1 adequacy require measured pump running/stall current. Layout leaves these connections unrouted (ratsnest) on purpose.
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TAPH research — PRELIMINARY — NOT FOR FABRICATION
Sheet: /Pump switch/
File: pump.kicad_sch
Title: TAPH carrier rev 0 — Pump switch
Size: A4
Date: 2026-11-07
Rev: 0—prelim
KiCad E.D.A. 10.0.7
Id: 4/8

SERVO DOMAIN (servo supply voltage/current UNVERIFIED; mechanical safe state UNRESOLVED)



No servo pulses are produced while IO25 is high−Z (R10). What the servo/blade does WITHOUT pulses, after supply loss, or on a fault is NOT established: the mechanical safe state is UNRESOLVED and is not claimed by this circuit.
Servo power is NOT taken from the logic regulator. Servo current path widths are BLOCKED pending servo stall current.

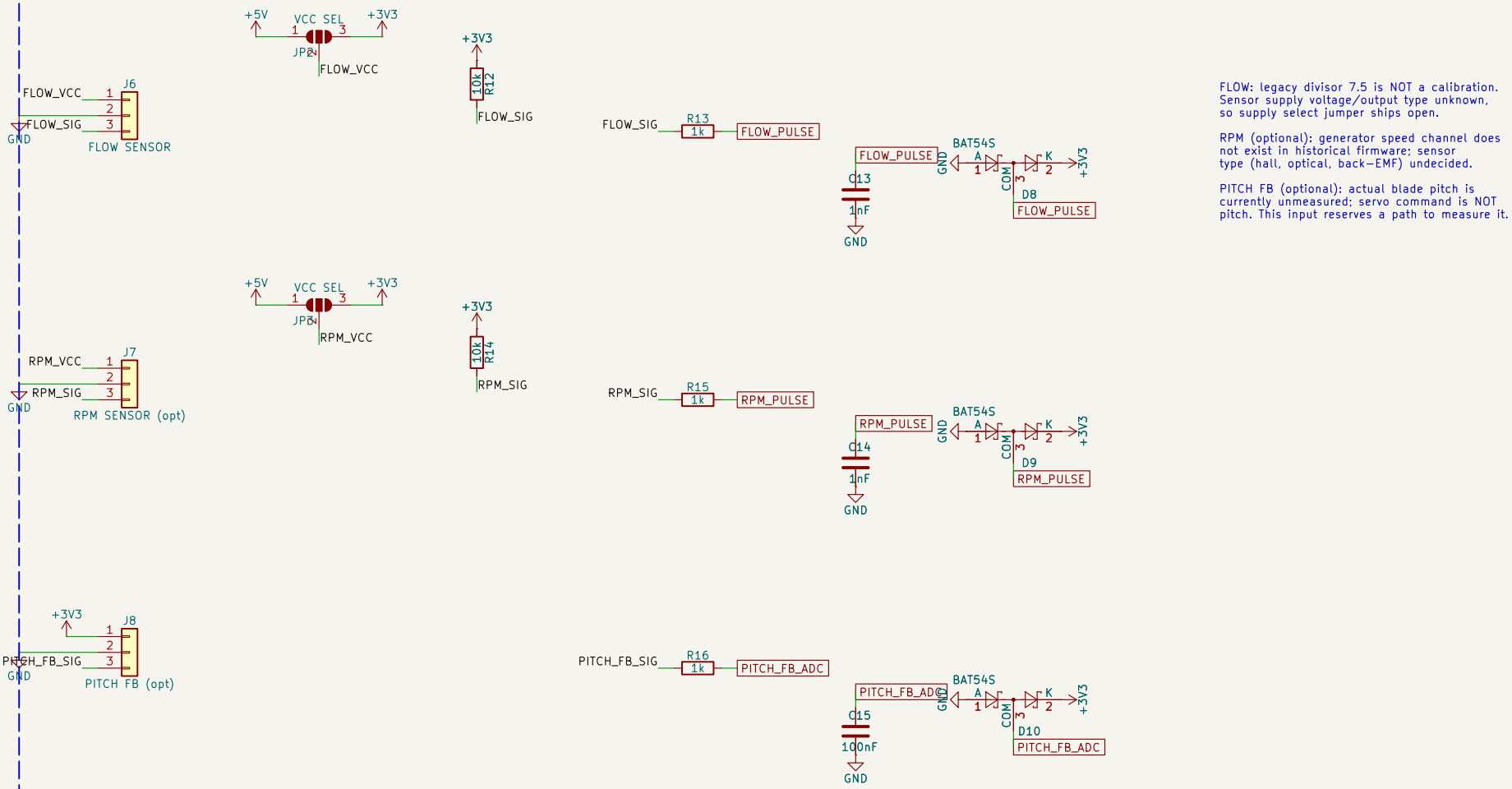
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TAPH research − PRELIMINARY − NOT FOR FABRICATION

Sheet: /Servo interface/
File: servo.kicad_sch

Title: TAPH carrier rev 0 − Servo interface

Size: A4	Date: 2026−11−07	**Rev: 0−prelim**
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EXTERNAL SENSOR INPUTS (sensor models, supply voltage and output type UNVERIFIED)

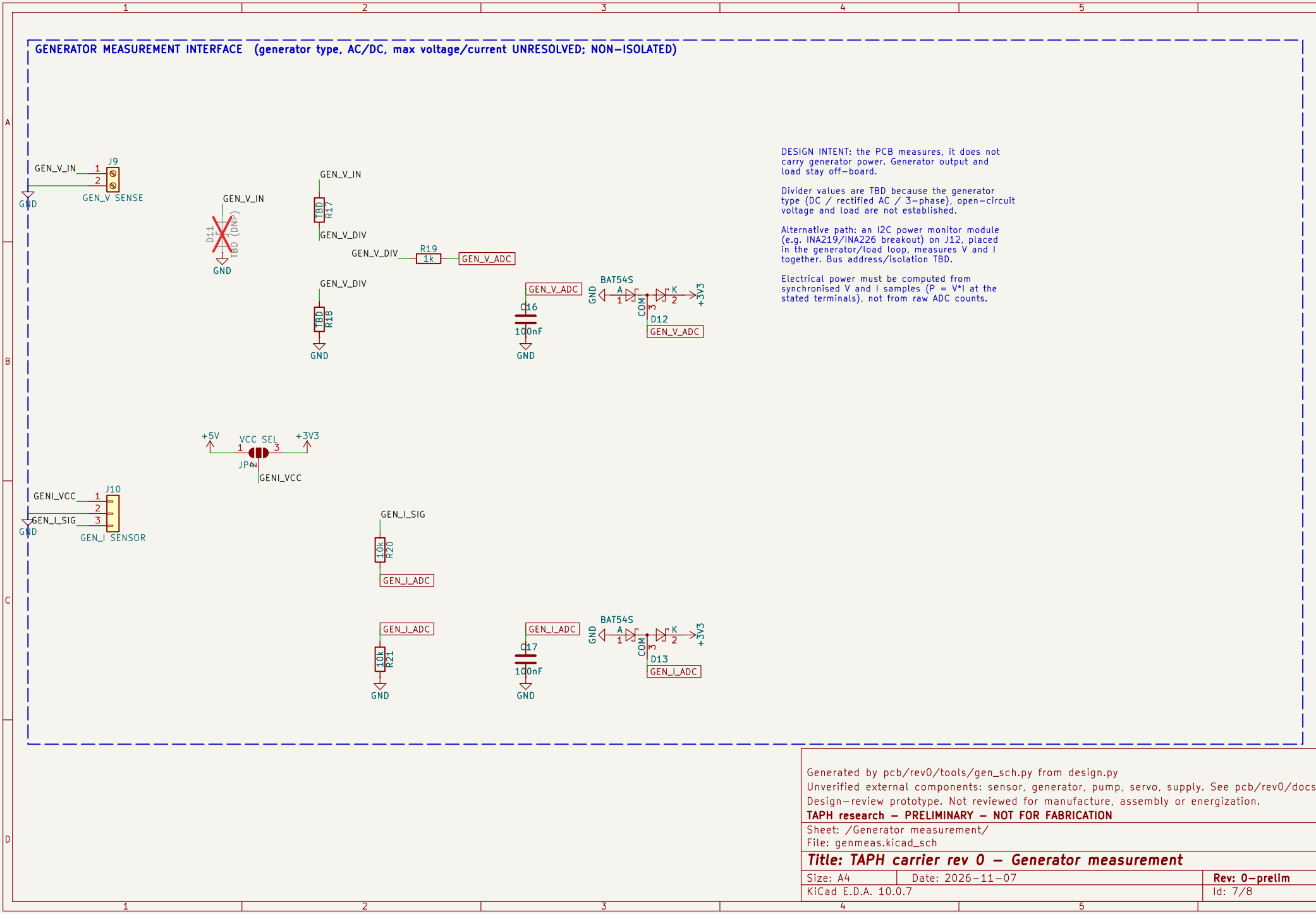


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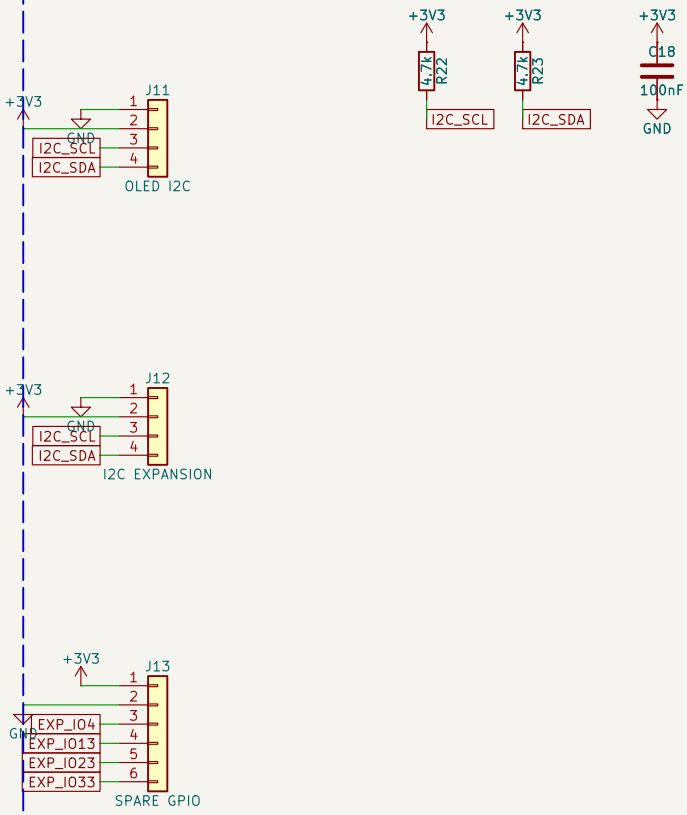
Sheet: /Sensor inputs/
File: sensors.kicad_sch

Title: TAPH carrier rev 0 — Sensor inputs

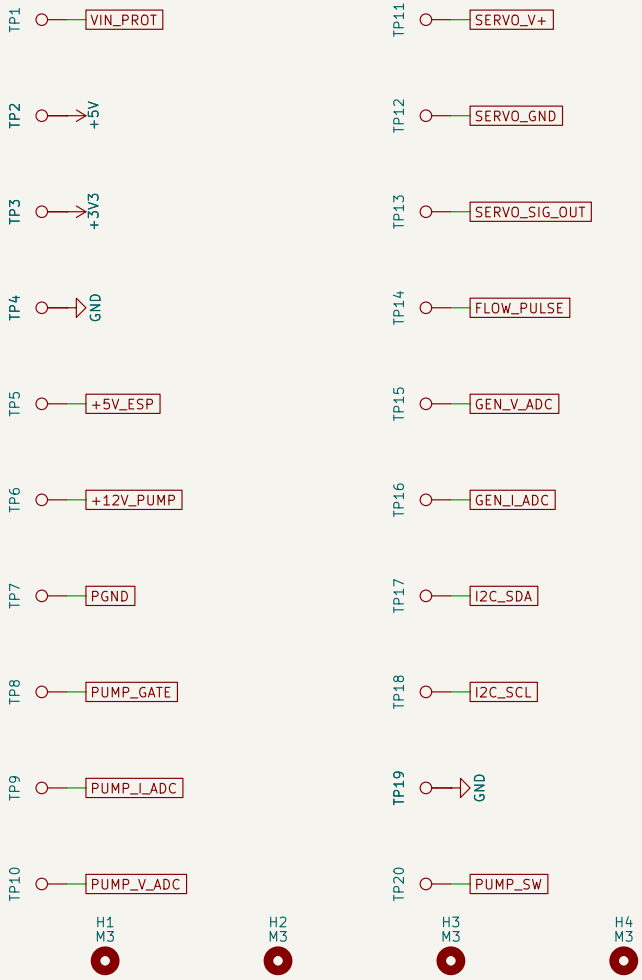
Size: A4	Date: 2026-11-07	**Rev: 0-prelim**
KiCad E.D.A. 10.0.7		Id: 6/8


GENERATOR MEASUREMENT INTERFACE (generator type, AC/DC, max voltage/current UNRESOLVED; NON-ISOLATED)
GEN_V_IN
J9
GEN_V SENSE
GND
D11
TBD (DNP)
R17
R18
R19
1k
GEN_V_DIV
GEN_V_ADC
C16
100nF
BAT54S
D12
+3V3
+5V
VCC SEL
JP4
GENI_VCC
J10
GEN_I SENSOR
GEN_I_SIG
R20
10k
R21
GEN_I_ADC
C17
D13
DESIGN INTENT: the PCB measures, it does not carry generator power. Generator output and load stay off-board.
Divider values are TBD because the generator type (DC / rectified AC / 3-phase), open-circuit voltage and load are not established.
Alternative path: an I2C power monitor module (e.g. INA219/INA226 breakout) on J12, placed in the generator/load loop, measures V and I together. Bus address/isolation TBD.
Electrical power must be computed from synchronised V and I samples (P = V*I at the stated terminals), not from raw ADC counts.
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TAPH research — PRELIMINARY — NOT FOR FABRICATION
Sheet: /Generator measurement/
File: genmeas.kicad_sch
Title: TAPH carrier rev 0 — Generator measurement
Size: A4
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Rev: 0-prelim
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Id: 7/8

OLED / I2C / EXPANSION



TEST POINTS & MECHANICAL



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Sheet: /HMI, expansion, test/
File: hmi.kicad_sch

Title: TAPH carrier rev 0 — HMI, expansion, test

Size: A4	Date: 2026-11-07	**Rev: 0-prelim**
KiCad E.D.A. 10.0.7		Id: 8/8